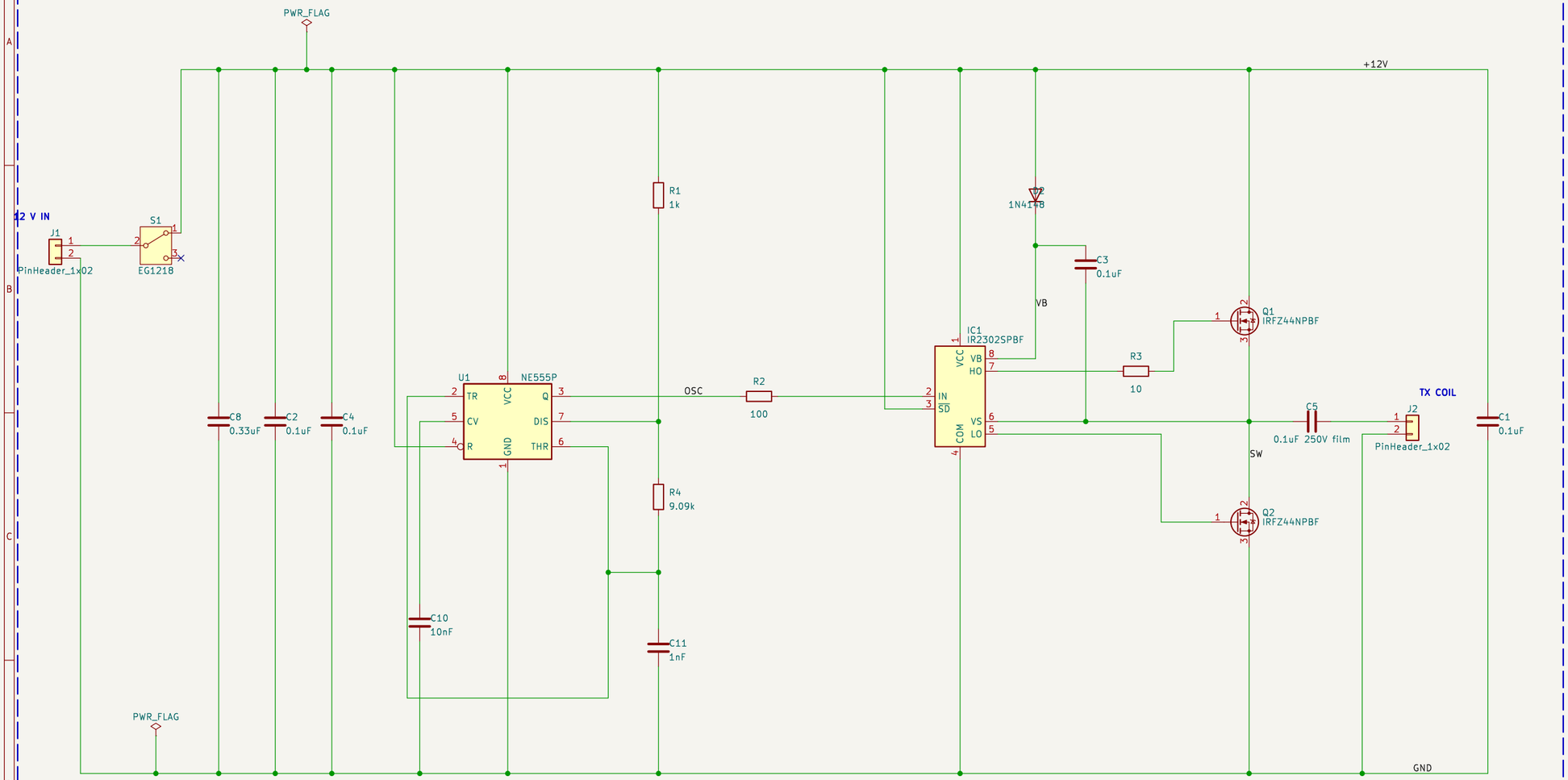
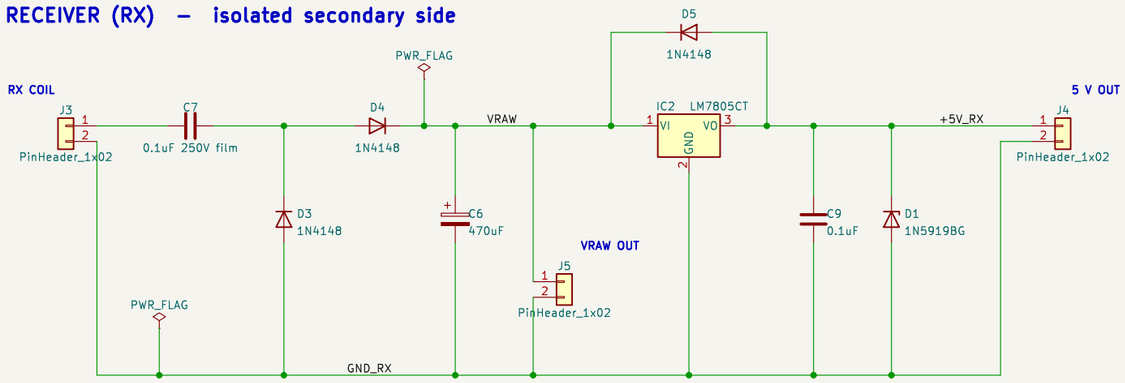
TRANSMITTER (TX) – 12 V DC primary side



RECEIVER (RX) – isolated secondary side



How it works
1. U1 (NE555) oscillates at f = 1.44/((R1+2*R4)*C11) = 75 kHz, ~53 % duty.
2. IC1 (IR2302) drives the Q1/Q2 half-bridge (built-in dead time, D2/C3 bootstrap).
3. The square wave at SW drives the series-resonant tank C5 + TX coil (J2).
4. RX coil (J3) + C7 resonate at the same 75 kHz; D3/D4 double the voltage into C6.
5. IC2 (LM7805) regulates to 5 V (J4). D5 protects IC2, D1 clamps over-voltage.
Coils: L = 1/((2*pi*75 kHz)^2 * 0.1 uF) = 45 uH each (~15 turns, 60 mm dia).
TX GND and GND_RX are NOT connected: energy crosses only via the magnetic field.
C5/C7: polypropylene film >= 250 V (>100 V at resonance). VRAW must stay above ~7.5 V for the 7805.

Fully wired: NE555 + IR2302 half-bridge transmitter, voltage-doubler + LM7805 receiver

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Title: Electromagnetic Coupling Link (Resonant Inductive TX/RX)

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